

Kitchen Sink

Every Feature in One File

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Abstract

This document exercises every supported feature of Calepin in a single file.

Keywords test, features

0.1 Markdown basics

This section tests standard CommonMark and GFM features.

0.1.1 Inline formatting

Italic, **bold**, ***bold italic***, ~~strikethrough~~, and `inline code`.

0.1.2 Lists

Ordered:

1. First item
2. Second item
3. Third item

Unordered:

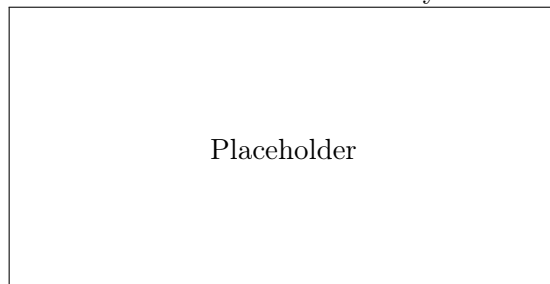
- Alpha
- Beta
- Gamma

0.1.3 Blockquote

This is a blockquote with **bold** and *italic* text.

0.1.4 Links and images

A link to nowhere and a reference-style link.



0.1.5 Footnotes

Here is a sentence with a footnote¹.

0.1.6 Horizontal rule

0.1.7 Table

Name	Score	Grade
Alice	95	A
Bob	87	B
Carol	92	A

0.2 Code chunks

0.2.1 R chunks

```
x <- 1:10  
mean(x)
```

```
> [1] 5.5
```

Echo fenced

¹This is the footnote content.

```
““{r, chunk-2}  
#| echo: fenced  
summary(mtcars$mpg)  
““
```

```
>      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.  
>  10.40   15.43   19.20   20.09   22.80   33.90
```

Figure from R

```
plot(1:10, (1:10)^2, pch = 19, col = "steelblue",  
     xlab = "x", ylab = "x squared")
```

See Figure 1 for the scatter plot.

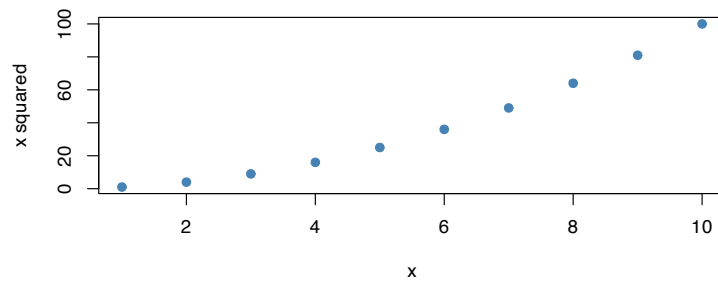


Figure 1: Scatter plot of x vs x^2

Echo false

Eval false

```
stop("this would error if evaluated")
```

Include false (silent execution)

Results asis

```
cat("**This is bold from asis output.**\n")
```

```
**This is bold from asis output.**
```

Warning and message control

```
warning("suppressed warning")  
message("suppressed message")  
cat("Only this output appears.\n")
```

```
> Only this output appears.
```

0.2.2 Python chunks

```
import math  
print(f"Pi is approximately {math.pi:.4f}")
```

```
> Pi is approximately 3.1416
```

```
squares = [x**2 for x in range(1, 6)]  
print(squares)
```

```
> [1, 4, 9, 16, 25]
```

0.2.3 Shell chunks

```
echo "Hello from shell"
```

```
> Hello from shell
```

0.2.4 Inline code

The mean of 1 to 10 is 5.5. Pi is 3.14. The secret value is 42.

0.3 Math

0.3.1 Inline math

The Pythagorean theorem states $a^2 + b^2 = c^2$.

A literal dollar sign costs \$5.

0.3.2 Display math

$$\int_0^1 x^2 dx = \frac{1}{3}$$

0.3.3 Labeled equation

$$E = mc^2 \tag{1}$$

0.3.4 Another labeled equation

$$\sum_{i=1}^n i = \frac{n(n+1)}{2} \tag{2}$$

See Equation 1 and Equation 2.

0.4 Cross-references

We can reference sections (Section 1, Section 3), figures (Figure 1), tables (Table 1), equations (Equation 1), theorems (Theorem 1), definitions (Definition 1), and code listings (Listing 1).

Bracketed form: [Figure 1]. Suppress type: 1. Grouped: [Equation 1; Equation 2].

0.5 Callouts

Note

0.5.1 A note

This is a **note** callout with a custom title.

Tip

A tip callout with the default title.

Warning

0.5.2 Watch out

A warning callout.

Important

An important callout with the default title.

Caution

0.5.3 Collapsible caution

This caution callout starts collapsed.

0.6 Theorems

Theorem 1.

0.6.1 Fermat's Last Theorem

For $n > 2$, there are no positive integers a, b, c such that $a^n + b^n = c^n$.

Proof. The proof was completed by Andrew Wiles in 1995 and is too long to include here.

□

Definition 1.

0.6.2 Group

A group is a set G with a binary operation satisfying closure, associativity, identity, and invertibility.

Lemma 1. *Every finite group has an element of order dividing the group order.*

Corollary 1. *Every group of prime order is cyclic.*

Example 1. The integers under addition form a group.

Remark 1. Groups are fundamental to abstract algebra.

By Theorem 1 and Definition 1, we see the connection. Also see Lemma 1, Corollary 1, Example 1, and Remark 1.

0.7 Tabsets

0.7.1 Tab A

Content of the first tab.

0.7.2 Tab B

Content of the second tab with some math: $e^{i\pi} + 1 = 0$.

0.7.3 Tab C

```
cat("Code in a tab\n")
```

```
> Code in a tab
```

0.8 Layouts

0.8.1 Two-column layout

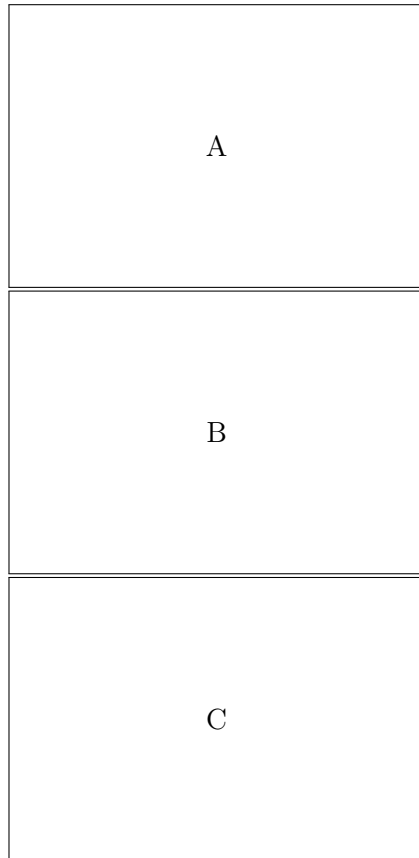
Left column content. This paragraph appears on the left side of a two-column layout.

Right column content. This paragraph appears on the right side.

Table 1: Number of known moons for selected planets.

Planet	Moons
Earth	1
Mars	2
Jupiter	95

0.8.2 Custom grid layout

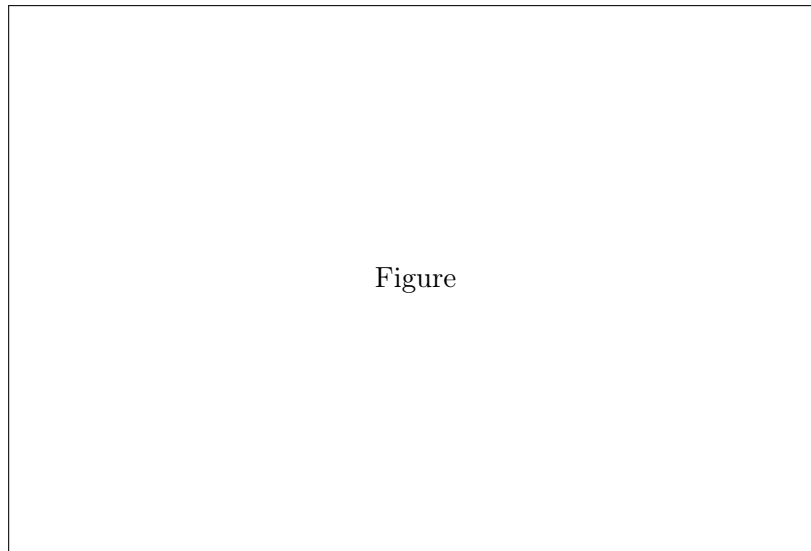


0.9 Figure divs

See Figure 1.

0.10 Table divs

See Table 1.



Figure

Figure 2: A figure div with a caption.

0.11 Code listings

```
square <- function(x) x^2  
square(7)
```

```
> [1] 49
```

See Listing 1.

0.12 Content visibility

This paragraph is only visible in **LaTeX** output.

This paragraph is hidden in HTML output but visible elsewhere.

This paragraph is visible because `show_extra` is true.

0.13 Hidden div

The hidden div executed: computed but not shown.

0.14 Raw blocks

This is raw **LaTeX**.

0.15 Jinja features

0.15.1 Context variables

The document title is: Kitchen Sink.

The sample size is: 50.

The current format is: latex.

0.15.2 Conditionals

Rendering to latex.

0.15.3 Loops

Features of this document:

- R code chunks
- Python code chunks
- Cross-references

0.15.4 Built-in spans

Lorem ipsum dolor sit amet consectetur adipiscing elit. Elit sed do eiusmod tempor incididunt ut labore et dolore magna.

After the page break.

0.16 Syntax highlighting

Non-executable code blocks with syntax highlighting:

```
function greet(name) {  
  return 'Hello, ${name}!';  
}
```

```
def factorial(n: int) -> int:  
  return 1 if n <= 1 else n * factorial(n - 1)
```